

Lab 4 — Reflection & Learning Summary

Dataset: Medical Insurance Costs | Tools: Python, statsmodels, Streamlit, Plotly

1. Understanding Dataset

Before running a single test, it is important to understand the numerical profile of the dataset— age, BMI, number of children, and charges. The shape of each variable decides which statistical tests are even valid later on.

Central tendency and spread

Mean and median describe the 'center' of a distribution but disagree sharply when data is skewed. In the charges column, the mean (roughly \$13,270 across the full dataset) sits well above the median, which is the single clearest early signal that a small number of very expensive claims are dragging the average upward while most patients pay far less.

Standard deviation quantifies the average distance of each point from the mean, but it treats every deviation the same way regardless of direction, and it's easily inflated by a handful of extreme values. The interquartile range (IQR = $Q3 - Q1$) sidesteps that problem by only looking at the middle 50% of the sorted data, so I used it as a cross-check whenever the standard deviation looked unusually large.

Skewness and kurtosis — why they mattered here

Skewness measures asymmetry. A skewness of roughly 1.5 for charges (the value the dashboard reports) means a pronounced right tail: most values cluster at the low end, with a long stretch of increasingly rare, increasingly expensive cases stretching out to the right. This single number is what justified, later on, choosing a non-parametric test over a t-test for that variable.

Kurtosis measures tail-weight relative to a normal curve, where a normal distribution has a kurtosis of 3 (or an 'excess kurtosis' of 0 depending on convention). A value noticeably above that threshold means the distribution produces more extreme outliers than a bell curve would predict — exactly the pattern visible in the charges histogram, where a cluster of low-cost non-smokers sits next to a separate, fatter-tailed cluster of high-cost smokers.

Practical lesson: none of these four numbers mean much in isolation. A histogram or KDE plot next to them is what actually reveals whether you're looking at one skewed population or two overlapping ones — which turned out to be the real story in this dataset.

2. Hypothesis Testing

Every test in Part 1 follows the same four-step skeleton, and getting this logic straight mattered more than memorizing any individual formula:

- Define a null hypothesis (H_0) — typically 'no difference' or 'no relationship' — and an alternative (H_1) that contradicts it.
- Choose a significance level α (the lab fixes this at 0.05), which is the probability of a false alarm we're willing to tolerate.

- Compute a test statistic from the sample data and convert it into a p-value — the probability of seeing a result this extreme (or more extreme) if H0 were actually true.
- Compare p to α : if $p < \alpha$, reject H0 in favor of H1; otherwise, fail to reject H0.

One subtlety that took some time to sit right: 'fail to reject H0' does not mean H0 is true — it only means the sample didn't provide strong enough evidence to overturn it. There may simply not have been enough data, or the true effect may be smaller than the test could detect.

The two ways a test can be wrong

	H0 is actually true	H0 is actually false
We reject H0	Type I error (α) — false positive	Correct decision
We fail to reject H0	Correct decision	Type II error (β) — false negative

Lowering α reduces the chance of a Type I error but raises the chance of a Type II error — the two trade off against each other, which is why 0.05 is a convention rather than a law.

3. Checking Assumptions Before Picking a Test

The choice of test has to be justified first by checking two assumptions on the groups being compared (smokers vs. non-smokers, in this case).

Shapiro-Wilk test for normality

This test asks whether a sample's ordered values line up with what you'd expect from a normal distribution. It produces a statistic W between 0 and 1 (values close to 1 support normality) and a p-value. A small p-value here is actually 'bad news' for a standard t-test, because it means the normality assumption behind that test has been violated. Running this separately on the smoker and non-smoker charge groups is important — one group can be reasonably normal while the other isn't.

$$W = \frac{(\sum_{i=1}^n a_i x_{(i)})^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

Levene's test for equal variance

Levene's test checks whether the two groups have similar spread, using deviations from each group's median rather than its mean (which makes it more robust to outliers than an F-test would be). This matters because the classic two-sample t-test assumes equal variances; if that assumption fails, a Welch-corrected t-test — which does not pool the variances — is the safer parametric option.

$$W = \frac{N - k}{k - 1} \cdot \frac{\sum_{i=1}^k N_i (\bar{Z}_{i\cdot} - \bar{Z}_{\cdot\cdot})^2}{\sum_{i=1}^k \sum_{j=1}^{N_i} (Z_{ij} - \bar{Z}_{i\cdot})^2}$$

What this meant for the insurance data

In this dataset, charges for both smokers and non-smokers failed the Shapiro-Wilk normality check (unsurprising, given the skew discussed in Section 1). That single result determined the entire downstream choice: since normality was violated, the correct test was not a t-test at all, but the non-parametric Mann-Whitney U test described next. This is a good example of how the 'boring' assumption checks actually drive the whole analysis rather than being a formality to skip.

4. Comparing Groups: Four Tests, Four Situations

Before conducting any analysis, understanding when to reach for each of the four distinct comparison tools is essential. Because each tool is designed for a specific combination of variable types and group counts.

Test	Compares	Use when
Two-sample t-test	Means of 2 groups	Continuous outcome, both groups roughly normal
Mann-Whitney U	Rank distributions of 2 groups	Continuous outcome, normality violated
Chi-square test	Two categorical variables	Testing association / independence
One-way ANOVA	Means of 3+ groups	Continuous outcome, roughly normal, equal variance

Two-sample t-test vs. Mann-Whitney U

The t-test statistic is essentially the difference between the two group means, scaled by how much variability we'd expect just from sampling noise — a large t-value means the observed gap is unlikely to be chance. The Mann-Whitney U test throws away the actual charge amounts and instead pools both groups, ranks every observation from lowest to highest, and asks whether one group's ranks are systematically higher than the other's. Because it works on ranks rather than raw values, a handful of extreme outliers can't distort the result the way they would in a mean-based test — which is exactly why it was the right choice once the normality check failed.

Chi-square test of independence

This test applies to two categorical variables — here, smoking status and region. It builds a contingency table of observed counts, then computes the counts you'd expect in each cell if the two variables were completely unrelated (using the row and column totals). The test statistic sums the squared, normalized gaps between observed and expected counts across every cell. A large statistic means the observed pattern deviates a lot from what independence would predict, pointing toward a real association rather than coincidence.

One-way ANOVA

ANOVA compares charges across all four regions at once by splitting the total variation in the data into two pieces: variation between the region means, and variation within each region (i.e., noise among individuals in the same region). The F-statistic is the ratio of these two pieces (scaled by their degrees of freedom); a large F means the differences between region averages are big compared to the natural spread inside each region, which is evidence that region matters.

A rejected ANOVA null only tells that at least one region differs from the others — not which one. Pinning that down would need a follow-up post-hoc test (e.g., Tukey's HSD), which was outside the scope of what this lab required.

5. Building the Regression Model

Part 2 shifts from comparing groups to modeling charges as a function of several predictors simultaneously — age, BMI, number of children, sex, smoker status, and region — fitted as an Ordinary Least Squares (OLS) regression in statsmodels.

What OLS is actually doing

OLS chooses the coefficients that minimize the total squared distance between the actual charges and the model's predicted charges — in other words, it finds the 'line' (or hyperplane, with multiple predictors) that fits the cloud of points as closely as possible in a least-squares sense. Categorical predictors like sex, smoker, and region can't be plugged in as text, so statsmodels internally converts them into 0/1 dummy variables (e.g., smoker_yes = 1 or 0), with one category held out as the baseline that the others are compared against.

output

- **Coefficients (β):** each one is the expected change in predicted charges for a one-unit increase in that predictor, holding every other predictor fixed. In this dataset the smoker coefficient dwarfs every other predictor — smoking status alone explains most of the spread in charges, which lines up with the boxplot in the dashboard's Data Exploration tab.
- **p-values:** tell you whether a coefficient is statistically distinguishable from zero, given the sample size and noise in the data. A predictor with a large p-value (like sex, in most cuts of this dataset) isn't pulling its weight in the model.
- **95% confidence intervals:** give a plausible range for the true coefficient rather than pretending the point estimate is exact — useful for communicating uncertainty rather than false precision.
- **R² and Adjusted R²:** R² is the fraction of the total variation in charges that the model explains. Adjusted R² applies a penalty for every additional predictor, so it only increases if a new variable genuinely improves the fit rather than just adding noise — which is the fairer number to quote once you start comparing models with different numbers of predictors.

6. Making Sure the Regression Can Be Trusted

A regression's coefficients, p-values, and confidence intervals are only valid if the model roughly satisfies the classical Gauss-Markov assumptions. Part 2 required a full diagnostic pass on the fitted model rather than accepting R² at face value.

Linearity and homoscedasticity

Checked with a residuals-vs-fitted plot: plot each residual (actual minus predicted) against the model's predicted value. If the relationship is genuinely linear and the error variance is constant, the points should scatter randomly around zero with no visible pattern. A funnel shape (residuals fanning out as fitted values grow) signals heteroscedasticity — the error variance is not constant — which biases the standard errors and makes the p-values from Section 5 unreliable.

Normality of residuals

A Q-Q (quantile-quantile) plot lines up the residuals' own quantiles against the quantiles you'd expect from a perfect normal distribution; points hugging the diagonal reference line support normality, while a curve away from it — especially in the tails — signals a violation. The Jarque-Bera test gives a numeric version of the same check, built from the residuals' own skewness and kurtosis. In this model, residuals showed a clear normality violation, almost certainly a knock-on effect of the raw charges variable being so right-skewed to begin with — a strong hint that fitting the model on $\log(\text{charges})$ instead of raw charges would likely fix a lot of this.

Multicollinearity via VIF

The Variance Inflation Factor measures how much a predictor's coefficient variance is inflated because that predictor can be predicted from the other predictors. It's computed by regressing each continuous predictor on all the others and using that regression's own R^2 ; a VIF above roughly 5 (sometimes 10) signals a problematic level of overlap. In this dataset, age, BMI, and number of children turned out to be fairly independent of one another, so VIF stayed comfortably low across the board — meaning the coefficients in Section 5 can be interpreted individually without worrying that they're just splitting credit for the same underlying signal.

Overall takeaway from diagnostics: a high R^2 is not proof of a good model. A model can explain plenty of variance and still fail the assumptions needed to trust its inference — which is exactly why these checks are a required step rather than an optional appendix.

7. The Dashboard: From Static Analysis to a Live Tool

Part 3 — wiring everything above into an app a user can actually manipulate. The deployed dashboard is organized into three tabs (Data Exploration, Hypothesis Testing Lab, and Live Prediction & Diagnostics), and the screenshots below are from the live build.

Live app: <https://202618006ds602-6nxns5wyrw35csdbdcokxf.streamlit.app/>

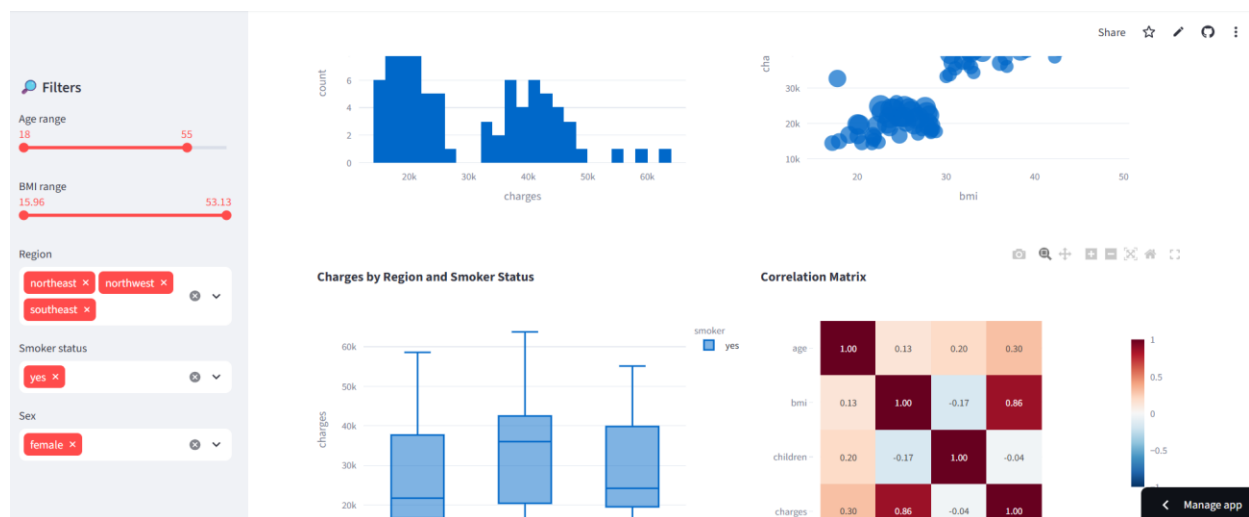
Sidebar filters drive everything downstream

The left-hand sidebar holds every filter — an age-range slider, a BMI-range slider, multi-select boxes for region and sex, and a smoker-status filter. In Streamlit, the whole script effectively reruns whenever any of these widgets change, so every statistic and plot on the page had to be written to recompute from the currently filtered subset rather than from the full dataset — a genuinely different way of writing code compared to a notebook, where a cell runs once and its output is frozen.



Data Exploration tab — header metrics and the live summary-statistics table, filtered to female smokers aged 18–55 in the northeast, northwest, and southeast regions.

With that specific filter combination applied (female, smoker = yes, three of the four regions, full age and BMI range), the dataset narrows from 1,338 records down to just 81. The header cards update instantly to reflect that subset: average charges jump to about \$29,897 and the median to \$24,536 — both far above the whole-dataset averages — which is exactly what you'd expect once the filter isolates only smokers, the single strongest driver of cost identified back in Section 5. The summary table underneath reproduces the descriptive statistics from Section 1 (count, mean, standard deviation, and so on) for age, BMI, children, and charges, but now computed live on whichever 81 rows match the sidebar's current selection.



Data Exploration tab — charges histogram, BMI-vs-charges scatter, a region/smoker boxplot, and the correlation heatmap, all recalculated for the filtered subset.

Four plots sit below the summary cards, and comparing them against the whole-dataset versions was informative in its own right:

- **Charges histogram:** even restricted to smokers only, the distribution is clearly bimodal rather than a single skewed hump — one cluster of smokers around \$15k–\$20k and a distinct second cluster above \$35k. That split lines up with the interaction the assignment specifically calls out: among smokers, BMI has a much

sharper effect on cost than it does for non-smokers, effectively creating two sub-populations within the 'smoker' group itself.

- **BMI-vs-charges scatter:** shows a visibly positive slope once the data is limited to smokers — higher BMI tracks with sharply higher charges. This is worth dwelling on, because in the unfiltered dataset the overall correlation between BMI and charges is fairly weak (routinely reported around 0.2); the relationship only becomes strong once you isolate smokers, which is a textbook example of an interaction effect hiding inside a simple correlation coefficient.
- **Boxplot by region and smoker status:** the three visible regions show broadly similar medians and spreads for this smoker-only subset, with the interquartile boxes overlapping heavily — visually consistent with the idea that, once you already know someone smokes, region adds comparatively little extra information about their charges.
- **Correlation heatmap:** for this filtered subset, BMI and charges show a correlation of 0.86 — dramatically higher than the same figure computed on the full dataset. Age and children stay weakly correlated with charges (around 0.30 and -0.04 respectively), reinforcing that smoker status and BMI are doing almost all of the explanatory work, which matches the dominant smoker coefficient from the regression in Section 5.

This is the single biggest practical lesson the dashboard taught me that the static notebook analysis didn't make as obvious: a correlation coefficient computed on the whole population can quietly hide a much stronger relationship inside a subgroup. Being able to slide the filters and watch the heatmap change in real time made the smoking–BMI interaction far more intuitive than reading a single static correlation matrix ever did.

Hypothesis Testing Lab tab

Rather than hard-coding 'smoker vs. non-smoker' as the only comparison, this tab exposes dropdowns for the grouping factor and the numeric metric, then automates the exact decision process from Section 3: it runs Shapiro-Wilk and Levene's test behind the scenes and silently routes to either a t-test or a Mann-Whitney U test depending on the result, before displaying the test statistic, p-value, and a plain-language Reject/Fail to Reject verdict at $\alpha = 0.05$. Turning that manual decision tree into code was a good forcing function for actually understanding the branching logic rather than just running whichever test the notebook happened to use.

Live Prediction & Diagnostics tab

This tab takes user-provided inputs — age and BMI sliders, a children stepper, and dropdowns for sex, smoker status, and region — feeds them into the fitted OLS model, and returns a point estimate for charges along with a 95% interval, refreshed on every input change. The residual diagnostic plots from Section 6 (residuals vs. fitted, Q-Q plot, and the VIF table) sit alongside it so a user can sanity-check the prediction against how trustworthy the underlying model actually is, rather than treating the number as gospel.

8. Overall Takeaways

- Descriptive statistics are a starting hypothesis, not a conclusion — they tell you what to expect, but the formal tests are what actually confirm or overturn that intuition with a defensible number.
- The choice between a t-test and Mann-Whitney (or between a plain OLS fit and a transformed one) isn't a style preference — it's dictated by the normality and variance checks that come before it.
- A regression's R^2 is not the finish line; the residual diagnostics decide whether its coefficients and p-values can be trusted at all.

- A correlation computed on an entire population can conceal a much stronger relationship living inside a subgroup — the dashboard's live filtering made the BMI $\times$ smoker interaction in this dataset far more obvious than a single static correlation matrix did.
- Turning a one-off analysis into an interactive tool forces a different mindset: every statistic has to stay correct under any possible filter combination a user might pick, not just the one combination that was tested first.

Overall, this lab connected the statistical theory from earlier coursework to something closer to how these techniques actually get used in practice — as a live, explorable decision-support tool rather than a static report that only answers the one question it was written to answer.